

Chapter :-1

INTRODUCTION

The Cafe Billing is a software where anyone can calculate and generate a billing receipt. The purpose of the system is to automate the calculation and efficiently generate the purchase bill digitally. The project is built with the use of tkinter module and mysql where it is composed of different widgets that basically use for designing the interface and save the billing data. The project can be openly-access without the use of any kind of login information. The user can do the inputting of quantity of the ordered items, he/she also needs to enter the name of the customer and phone number. After the order is done the user will click the generate bill to display the receipt of the purchase item. It also saves the customer's purchasing details using the mysql database system. It also displays the saved billing information of all the customers. This software project consists of an effective and easy user interface to help the employee in easy bill and calculation providing an efficient customer services

It covers major aspects of billing, it could perform the following operations :-

- Reducing long queues
- Saving time
- Track billing
- Prepare receipt
- Cost free
- Enables 24*7 service
- Manage previous record

Chapter :-2

LITERATURE SURVEY AND APPLICATION

Modern retail and food industries have

shifted towards POS (Point of Sale) systems. Research shows that businesses using automated billing see a 40% increase in operational efficiency. We studied existing models like the ones used in major food chains (Starbucks/CCD) to understand the workflow of order management and inventory tracking.

2.2 Applications

- Small Cafes & Bistros: Ideal for outlets with limited staff.
- University Canteens: Can handle high volumes of students quickly.
- Bakery Shops: Useful for tracking daily sales of perishable items.
- Fast Food Chains: Facilitates quick order-to-delivery cycles.

The main objective of the Cafe Billing is to manage the details of Payments and Salary. It manages all the information about the employees of the organization and their payroll.

The main objectives of Cafe Billing are:

- Provides the searching facilities of a particular employee.
- The transactions are executed in offline mode.
- Manage the information of billing.
- It enables easier billing transactions.

2.1 Hardware and Software Specifications

HARDWARE:

- Processor Intel(R) Core(TM) i3-1005G1 CPU @ 1.20GHz 1.19 GHz
- Installed RAM 12.0 GB (11.8 GB usable)
- HDD 1tb

SOFTWARE:

- Front End-Python 3.6 or above
- Back End-MYSQL server 5.0 or above

Chapter :- 3

THEORY BEHIND

What is Python?

Python is a popular programming language. It was created by Guido van Rossum, and released in 1991.

It is used for :

- Web development(server-side),
- Software development,
- Data analysis,
- System scripting

What Python can do?

- Python can be used on a server to create web application.
- Python can be used alongside software to create workflows.
- Python can connect to database system. It can also read and modify files.
- Python can be used to handle big data and perform complex mathematics.
- Python can be used for rapid prototyping, or for production-ready software development.

- **What is MySQL?**

MySQL is an open-source relational database management system(RDBMS). It is the most popular database system used with PHP. MySQL is developed, distributed, and supported by Oracle corporation.

- The data in a MySQL database are stored in tables which consist of column and rows.
- MySQL is a database system that runs on a server.
- MySQL is ideal for both small and large application
- MySQL is very fast, reliable and easy to use database system. It uses standard SQL.
- MySQL compiles on a number of platforms.

Interface Python with MySQL

MySQL Python/Connector is an interface for connecting to a MySQL database server from Python. It implements the Python database API and is built on top of the MySQL.

The general workflow of a Python program that interacts with a MySQL-based database is as follows :

- Connect to the MySQL server.
- Create a new database.
- Connect to the newly connected or an existing database.
- Execute a SQL query and fetch results.
- Inform the database if any changes are made to a table.
- Close the connection to the MySQL server.

Functions and Modules used in project

- **Mysql.connector**

It is used to connect mysql databases with python.

- **Time and date**

It is used to generate current date and time

Chapter :- 4

EXPERIMENT SIMULATATION AND TESTING

Database:-

Name of database :- cafe
Number of tables :- 1
Name of tables :- cafe_data

Tables:-

1.cafe_data:-

```
mysql> desc cafe_data;
```

Field	Type	Null	Key	Default	Extra
order_no	int	YES		NULL	
customername	varchar(30)	YES		NULL	
item_name	varchar(50)	YES		NULL	
date	date	YES		NULL	
visiting_time	varchar(20)	YES		NULL	
phone_no	int	YES		NULL	

6 rows in set (0.01 sec)

Table:- 1.0

Source Code

```
import mysql.connector
a= mysql.connector.connect(
    host = "localhost",
    user = "root",
    password = "alen",
    database = "cafe")
cur= a.cursor()
q="1"
while q=="1":
    def mainf():

print("#####
#####")

    print( "
INSPIRON cafe
")

print("#####
#####")

    def pencil():
        import mysql.connector
        a= mysql.connector.connect(
            host = "localhost",
            user = "root",
            password = "alen",
            database = "cafe")
        cur= a.cursor()
```

```

        data= "insert into cafe_data
values (%s,%s,%s,%s,%s,%s) "
        cur.execute(data,r)
        a.commit()
#data
from datetime import date
today = date.today()

#current time
from datetime import datetime
now = datetime.now()
current_time = now.strftime("%H:
%M:%S")
#serial number function
def num():
    for oo in range(100):
        a=None
    return oo
def access():
    import mysql.connector
    a= mysql.connector.connect(
        host = "localhost",
        user = "root",
        password = "alen",
        database = "cafe")
    cur= a.cursor()
    cur.execute(kkk)
    data=cur.fetchall()
    for i in data:
        print(i)

```



```
dino={1:"coldcoffe",2:"Burger",3:"Pizza",4:
"Cold Drink",5:"Pasta",6:"Chilli Potato"
```

```
,7:"Tea",8:"Chowmin",9:"Fried rice"}
```

```
dinoprice={1:40,2:50,3:250,4:25,5:60,6:7
0
,7:20,8:40,9:50}
```

```
price=0
print("""[1]menu
[2]old records
[3]staff members info
[4]exit """)
```

```
orderlist=[]
qtylist=[]
n=int(input("enter your
option:"))
if n==1:
```

```
print("*****MENU*****")
```

```
print("""
[1]cold coffee      rs40
[2]Burger           rs50
[3]pizza            rs250
[4]cold Drink       rs25
[5]Pasta            rs60
```

```

[6]Chilli potato    rs70
[7]Tea              rs20
[8]Chowmin          rs40
[9]Fried Rice       rs50
    """
    choice="y"
    while choice=="y":
        order=int(input("enter
the items no. to order:"))
        qty=int(input("enter the
quantity:"))
        qtylist.append(qty)

    price+=qty*dinoprice[order]

    orderlist.append(dino[order])
    choice=input("press y to
add more items or n to go further:")

    QWE=input("enter y to
confirm your order or press enter to
exit:")

    if QWE=="y":
        name= input("enter your
name: ")

        phone=int(input("enter
your phone number:"))

    print("""

```

***** INSPIRON CAFE

BILL

""")

print(today,current_time)

print("name:",name)

print("phone:",phone)

print("----your
order----")

for i in
range(len(orderlist)):

print(orderlist[i],qtylist[i])

print()
print("total bill",
price)

print("*****
*****")

r=(num(),name,str(orderlist),today,current_time,phone)

pencil()

print()

```
        print("bill generated  
successfully")
```

```
        print()
```

```
print("@@@@@@@@@@@@thankyou for  
visiting.!@@@@@@@@")
```

```
        print()
```

```
        print()
```

```
    else:
```

```
        print("thankyou")
```

```
    elif n==2:
```

```
        print("$$$$$$$$$$$$$$$$$$$$$$$$$$$$$$  
$ OLD RECORDS $$$$$$$$$$$$$$$$$$$$$$$$$$$$$$  
$$$$$")
```

```
        ab=input("eter order_no to  
access the old records:-")
```

```
        cur.execute("select * from  
cafe_data where order_no={}".format(ab))  
        data=cur.fetchall()
```

```
        for row in data:  
            print("Serial  
no.:",row[0])  
            print("Customer  
name:",row[1])  
            print("customer phone  
no.:",row[5])
```

```
print("Items:",row[2])
print("Date:",row[3])
print("Time",row[4])
```

```
elif n==3:
```

```
print("-----")
    print("our staff members
are:-")
        print("""(1) vishwajeet
maurya (Manager)
(2) Ankit Gupta
(3) pushpam jha vatsal """)
```

```
elif n==4:
```

```
    print("thankyou")
```

```
else :
```

```
    print("wrong choice")
```

```
mainf()
```

```
q=input("do you want to run it in
continue:(press 1)")
```

Outputs

● Main Menu

```
===== RESTART: C:\Users\ankit\Desktop\cafetry.py =====
#####
                        INSPIRON cafe
#####
[1]menu
[2]old records
[3]staff members info
[4]exit
enter your option:
```

Menue fig 1.0

● Menu

```
===== RESTART: C:\Users\ankit\Desktop\cafetry.py =====
#####
                        INSPIRON cafe
#####
[1]menu
[2]old records
[3]staff members info
[4]exit
enter your option:1
*****MENU*****

[1]cold coffee      rs40
[2]Burger           rs50
[3]pizza            rs250
[4]cold Drink       rs25
[5]Pasta            rs60
[6]Chilli potato    rs70
[7]Tea              rs20
[8]Chowmin          rs40
[9]Fried Rice       rs50

enter the items no. to order:
```

Ordering fig 1.1

● Ordering

*****MENU*****

[1]cold coffee	rs40
[2]Burger	rs50
[3]pizza	rs250
[4]cold Drink	rs25
[5]Pasta	rs60
[6]Chilli potato	rs70
[7]Tea	rs20
[8]Chowmin	rs40
[9]Fried Rice	rs50

enter the items no. to order:2
enter the quantity:3
press y to add more items or n to go further:y
enter the items no. to order:3
enter the quantity:1
press y to add more items or n to go further:n
enter y to confirm your order or press enter to exit:y
enter your name: ankit
enter your phone number:783697656

Bill fig 1.2

✓Bill

***** INSPIRON CAFE *****
BILL

2022-12-07 22:42:21

name: ankit

phone: 783697656

----your order----

Burger 3

Pizza 1

total bill 400

bill generated succesfully

@@@@@@@@@@@@thankyou for visiting.@@@@@@@@@@@@

● Old record.

Old record fig 1.3

```
===== RESTART: C:\Users\ankit\Desktop\cafetry.py =====
#####
                        INSPIRON cafe
#####
[1]menu
[2]old records
[3]staff members info
[4]exit
enter your option:2
$$$$$$$$$$$$$$$$$$$$ OLD RECORDS $$$$$$$$$$$$$$$$$$$$
enter order_no to access the old records:-101
Serial no.: 101
Customer name: ankit
customer phone no.: 783697656
Items: ['Burger', 'Pizza']
Date: 2022-12-07
Time 22:42:21
|
```

Staff member info fig 1.4

● Staff member info

```
===== RESTART: C:\Users\ankit\Desktop\cafetry.py =====
#####
                        INSPIRON cafe
#####
[1]menu
[2]old records
[3]staff members info
[4]exit
enter your option:3
-----
our staff members are:-
(1) ankit gupta
(2) vishwajeet maurya
(3) pushpam jha vatsal
|
```


Chapter :- 5

CONCLUSION LIMITATION AND FURTHER WORK

- You can access the old records only through order number.
- The transactions are executed in off-line mood , hence on-line data for payment, discount captures and modification are not possible.
- It does not include the management of stocks (of food items) , it only include the billing system.

Conclusion

Cafe Billing System software developed for a restaurant has been designed to achieve maximum efficiency and reduce the time taken to handle the billing activity. It is designed to replace an existing manual record system thereby reducing time taken for calculations and for storing data.

Bibliography

To develop this project many references were used :

- Computer Science with Python by Sumita Arora,Dhanpat Rai Publications.
- <https://www.google.com>
- <https://www.w3schools.com>
- <https://www.python.org.in>
- <https://www.mysql.org>